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OF COLLOIDS
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MPI of Colloids and Interfaces · Am Muehlenberg 1 · D-14476 Potsdam

Dr. Felix Loeffler
MPI of Colloids and Interfaces
Biomolecular Systems Department
Am Muehlenberg 1
D-14476 Potsdam
+49 331 567-9359
http://www.mpikg.mpg.de/SAT
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To whom it may concern,

I am Dr. Felix Loeffler, and I am writing to you to extend my professional backing for Dr. Akul Yugesh Mehta. Although I have never worked with Dr. Mehta, we share a number of research interests. I am thus very familiar with his research and offer a unique and impartial perspective of his contributions to the field of biotechnology. At this time, I am a group leader at the Max Planck Institute of Colloids and Interfaces where I have worked since 2017. Prior to this, I was a postdoctoral researcher at the Karlsruhe Institute of Technology and a visiting scholar at the University of California Berkley. These experiences, as well as my doctorate in physics and biochemistry from the DKFZ German Cancer Research Center, allow me to expertly address Dr. Mehta's research and its many merits.

According to his CV, Dr. Mehta is an instructor of surgery and a staff scientist at the Beth Israel Deaconess Medical Center, Harvard Medical School, which follows his previous position there as a research scholar in the Department of Surgery. Academically speaking, Dr. Mehta completed a bachelor's degree in pharmaceutical sciences from the University of Mumbai, and a doctorate in pharmaceutical sciences from Virginia Commonwealth University, where he also worked as a postdoctoral scholar. Through these academic and professional experiences, Dr. Mehta has developed a strong skillset that allows him to excel in the field of biotechnology.

Additionally, Dr. Mehta stands as a leading contributor to the biotechnology field, evidenced by his position as a peer reviewer for journals like *Chemical Communications, RSC Advances,* and the *New Journal of Chemistry*. Additionally, he served as an associate editor for *Frontiers in Molecular Biosciences*. Major journals such as these only invite top-tier experts in the field to review and judge the work of their peers, clearly signifying that Dr. Mehta is seen as an authority in biotechnology research. Dr. Mehta has also helped to organize several international conferences, such as the 13th Jenner Glycobiology and Medicine Symposium (2019) and the 12th International Symposium on Glycosyltransferases – GlycoT 2020 (2020).

Due to the need to develop synthetic glycopeptides, which are useful when studying glycan interactions with other biomolecules, Dr. Mehta created a new technique that he called Glyco-SPOT synthesis. Previously, glycopeptide synthesis required many reagents, which increases the cost of producing large libraries of such molecules. The Glyco-SPOT technique allowed for the creation of glycopeptide libraries at nanomole quantities using cellulose filter paper as a solid support. Dr. Mehta's method used up to ten times less glycoamino acids than traditional synthesis techniques. His technique is also useful for the synthesis of libraries of other fragile post-translationally modified peptides such as those containing sulfotyrosines. His technique demonstrated the capability of simultaneously synthesizing 20 to 60 peptides at

a time. To date, no other researchers have produced such a high output of diverse glycopeptides as Dr. Mehta. Furthermore, Dr. Mehta's technique is compatible with microarray technology (which represents another area of his impressive expertise), making it a very efficient high-throughput screening platform. Using Glyco-SPOT synthesis with microarrays, Dr. Mehta created three microarrays that together contained over 120 different glycopeptides. Subsequently, these microarrays are being used by Dr. Mehta and other experts who are studying various antibodies and proteins, and characterizing their binding mechanisms.

The influence of this study was staggering, illustrated by my team's use of it in our 2022 paper in the *Journal of the American Chemical Society*. In this paper, we designed a method of producing glycans (precursors to glycopeptides), inspired by Dr. Mehta's results as a foundation to build from. Our use of his work shows the importance of his work among his peers in the field.

Glycopeptides are parts of glycoproteins and a vital component in numerous cellular processes, as they are found on the surface of a cell which allows them to influence how cells interact with each other and their environment. Not only is his Glyco-SPOT synthesis technique substantially more efficient than existing methods for the small-scale synthesis of glycopeptide libraries, but it is also much more cost-effective and requires no special equipment. His technique enables researchers around the world to synthesize glycopeptides and produce libraries of molecules more easily. As this process allows for the discovery of new biological pathways and the screening for new antibodies and therapies, Dr. Mehta's research is making the work of others in these important fields significantly easier. From his creation of Glyco-SPOT and glycopeptide microarrays, Dr. Mehta's peers are now able to discover drug targets, diagnostic targets, and therapies for a host of different diseases and serious illnesses. For these reasons, Dr. Mehta's research in glycopeptides and microarrays enables the success of researchers worldwide.

Indeed, Dr. Mehta has advanced the field of biotechnology through his capacity to synthesize glycopeptides and glycosulfopeptide libraries. Dr. Mehta's prowess in creating the state-of-the-art tool Glyco-SPOT and his many invitations to perform peer reviews for leading publications situate him as a leader in the field and establish his inherent value to the glycoscience community and the greater field of biotechnology in the United States and the world.

Sincerely,

Dr. Felix Loeffler

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